

1.1 Introduction

Green Market is a web-based agricultural marketplace and advisory platform designed to connect farmers, buyers, agricultural experts, and administrators within one secure digital ecosystem. The system responds to common challenges in the agricultural value chain, especially limited market access for farmers, fragmented communication between stakeholders, and difficulty in finding reliable agricultural support.

The current project is implemented as a full-stack Next.js application with role-based access for sellers, buyers, experts, and administrators. It supports product listing, order management, buyer shopping workflows, expert service publishing, consultation requests, reviews, notifications, messaging, and dashboard-based monitoring. These features allow agricultural stakeholders to interact through a structured platform instead of relying only on informal channels or middlemen.

The platform is built around three practical pillars. The first is a digital marketplace where sellers can publish farm products with descriptions, categories, images, prices, units of measure, stock quantities, and product status. The second is an expert service module where agricultural professionals can offer paid consultation services and manage requests from users who need technical guidance. The third is a management and accountability layer that includes user profiles, role-based dashboards, order fulfilment tracking, reviews, notifications, conversations, and administrative oversight.

By combining these modules, Green Market aims to improve transparency, reduce communication barriers, support direct agricultural trade, and make expert assistance more accessible. The system therefore contributes to the digital transformation of agriculture by providing a centralized platform for market participation, consultation, and operational management.

1.2 Problem Statement

Agriculture remains a major contributor to food security, employment, and economic development in Ghana and other developing economies. Despite its importance, many farmers still face difficulty reaching reliable buyers, presenting their products beyond local markets, and receiving fair value for their produce. Traditional market structures often depend heavily on intermediaries, which can reduce farmer earnings and limit price transparency.

Buyers also experience challenges when searching for fresh agricultural products from trusted sellers. The absence of a centralized digital marketplace can make it difficult to compare products, confirm availability, communicate with sellers, and track orders efficiently. This creates uncertainty in pricing, product quality, and supply reliability.

Another major challenge is limited access to agricultural expertise. Farmers and other agricultural stakeholders may require guidance on crop management, animal health, pest control, soil improvement, and general farm practices, but professional support is not always easily available. Where advice is available, communication and appointment management may still be informal and difficult to track.

There is also a need for better digital coordination within agricultural platforms. User registration, role management, product management, consultation booking, order processing, reviews, messaging, notifications, and administrative monitoring are often handled separately or manually. This fragmentation reduces efficiency and makes it harder to build trust among stakeholders.

Therefore, there is a need for an integrated web-based system that brings agricultural marketplace activities, expert consultation services, communication tools, and administrative controls into one platform. Green Market is proposed to address these gaps by improving direct interaction, transparency, accessibility, and digital management across the agricultural value chain.

1.3 Aim of the Project

The aim of this project is to design and develop Green Market, a secure and responsive web-based agricultural platform that enables farmers and sellers to market agricultural products directly to buyers while also providing access to expert consultation services and role-based management tools.

The project seeks to improve agricultural trade and support services by integrating product listing, order management, expert service booking, messaging, reviews, notifications, user profile management, and administrative oversight into a single digital platform. Through this integration, the system supports more transparent transactions, better communication, and more organized agricultural service delivery.

1.4 Objectives of the Project

The objectives of the Green Market project are as follows:

To design and develop a user-friendly digital marketplace where farmers and agricultural sellers can create, update, and manage product listings with relevant product details, images, prices, units of measure, stock quantities, and availability status.

To provide buyers with a convenient platform for browsing agricultural products, managing cart activities, placing orders, and tracking purchase-related information.

To implement role-based access control for buyers, sellers, experts, and administrators so that each user category can access the functions relevant to its responsibilities.

To develop an expert services module that allows agricultural experts to publish services, receive consultation requests, manage appointments, and provide guidance to platform users.

To support communication and trust through conversations, notifications, reviews, seller verification status, and administrative monitoring.

To build a secure, scalable, and responsive application using modern web technologies including Next.js, TypeScript, PostgreSQL, Drizzle ORM, Better Auth, Tailwind CSS, and Supabase storage support.

1.5 Core Features of the System

1.5.1 Digital Marketplace

The marketplace is the central trading component of Green Market. It allows sellers to list agricultural products with details such as title, description, category, price, unit of measure, stock quantity, image, and active status. Buyers can access the marketplace to view available products and begin purchase activities through the cart and order flow.

The system also supports order records, fulfilment status, shipping address information, and tracking number fields. These features help organize transactions and provide a clearer record of buyer and seller activities.

1.5.2 Expert Services and Consultation Requests

The expert services module allows agricultural professionals to create service listings with titles, descriptions, prices, and consultation durations. Users can request consultations, include messages, and schedule sessions. The system also supports consultation status, meeting provider, meeting link, and meeting notes, making expert interactions easier to manage.

This feature helps bridge the knowledge gap in agriculture by making professional guidance more accessible through the platform. It also creates a structured way for experts to manage their services and consultation history.

1.5.3 User Profiles and Role-Based Dashboards

Green Market provides different profile structures for sellers, buyers, and experts. Sellers can record farm-related information such as farm name, location, and verification status. Buyers can provide business information, while experts can indicate their expertise and years of experience.

The system includes dashboards for sellers, buyers, experts, and administrators. These dashboards organize user-specific activities such as product management, order monitoring, consultation management, analytics pages, profile setup, and system administration.

1.5.4 Communication, Reviews, and Notifications

To promote trust and collaboration, the platform includes conversations, messages, notifications, and reviews. These features support direct communication among users, help users receive timely updates, and allow participants to provide feedback on marketplace interactions.

1.5.5 Administration and System Monitoring

Administrative functions allow authorized users to monitor platform activity, manage users, review sellers, and oversee products. This improves platform governance and helps maintain reliability, accountability, and order within the system.

1.6 Target Users

The Green Market platform is designed for the following major user groups:

Farmers and sellers: These users list agricultural products, manage stock information, monitor orders, update seller profiles, and interact with buyers through the platform.

Buyers: Buyers browse products, manage cart activities, place orders, track purchases, and review their marketplace experience.

Agricultural experts: Experts publish consultation services, receive requests, manage schedules, and provide professional agricultural advice.

Administrators: Administrators monitor users, sellers, products, and general system activity to support secure and reliable platform operation.

1.7 Value Proposition

Green Market creates value by providing a centralized platform for agricultural trade, consultation, and communication. For farmers and sellers, the system increases product visibility and supports direct access to buyers. This can reduce overdependence on intermediaries and improve opportunities for fairer pricing.

For buyers, the platform improves access to agricultural products by allowing them to view listed items, compare available options, and place orders through a structured digital process. This supports convenience, transparency, and better supply discovery.

For agricultural experts, Green Market provides a digital channel for offering professional services, managing consultation requests, and reaching users who need guidance. For administrators, the system provides oversight tools that help maintain trust, security, and proper platform governance.

Overall, the platform improves market access, expert accessibility, communication, accountability, and digital organization within the agricultural value chain.

1.8 Scope of the System

The scope of this project covers the design and development of a web-based agricultural platform with marketplace, expert consultation, profile management, communication, review, notification, order management, and administrative features. The system is implemented as a responsive full-stack web application using Next.js, TypeScript, PostgreSQL, Drizzle ORM, Better Auth, Tailwind CSS, and Supabase storage support for uploaded images.

The system includes authentication, session management, and role-based access control for buyers, sellers, experts, and administrators. It also includes seller product management, buyer shopping

workflows, expert service management, consultation request handling, messaging, notifications, reviews, and order fulfilment tracking.

The project does not cover physical logistics such as transportation, warehousing, delivery fleet management, or direct farm operations. It also does not claim to fully automate agricultural decision-making. Instead, its focus is on digital market coordination, expert access, transaction organization, and stakeholder communication.

1.9 Project Deliverables

Upon completion, the project will produce the following deliverables:

A functional Green Market web application with marketplace, expert service, consultation, order, messaging, review, notification, dashboard, and administrative modules.

The complete source code of the system, including frontend pages, backend API routes, database schemas, migrations, services, validation logic, and supporting scripts.

A PostgreSQL database design managed through Drizzle ORM, covering users, profiles, products, expert services, consultation requests, orders, order items, reviews, conversations, messages, notifications, and administrators.

Project documentation explaining the problem, objectives, methodology, system design, implementation, testing, and conclusions.

A user guide or demonstration material showing how buyers, sellers, experts, and administrators interact with the platform.

1.10 Conclusion of Chapter

This chapter has introduced the Green Market project by presenting its background, problem statement, aim, objectives, core features, target users, value proposition, scope, and expected deliverables. The chapter establishes the need for a centralized agricultural platform that supports direct marketplace interaction, expert consultation, structured communication, and role-based system management.

The proposed system is relevant because it addresses practical challenges faced by agricultural stakeholders, including limited market access, fragmented communication, difficulty in accessing expert advice, and weak digital coordination. The next chapters will review related systems, explain the methodology, and present the design and implementation of the Green Market platform.

CHAPTER TWO

REVIEW OF RELATED WORKS / REVIEW OF SIMILAR SYSTEMS.

2.1 Introduction

This chapter presents a comprehensive review of related systems, technologies, and existing research relevant to the development of the Green Market platform. The purpose of this review is to examine current agricultural and digital marketplace solutions, analyze their operational processes, and identify the strengths and limitations that influence the design of the proposed system. By studying existing systems, the project establishes a strong foundation for understanding the technological gaps and practical challenges within the agricultural sector that Green Market seeks to address.

The chapter explores various agricultural technology platforms, online marketplaces, and digital service systems that provide functionalities such as product listing, online transactions, farmer support services, expert consultation, and data management. Particular attention is given to how these systems facilitate communication and interaction among stakeholders within the agricultural value chain, including farmers, buyers, and agricultural professionals. The review also evaluates the efficiency, usability, scalability, and technological capabilities of these existing platforms.

In addition, this chapter discusses the processes and operational workflows of existing systems, including their key features, advantages, and shortcomings. This analysis helps to identify critical limitations such as fragmented agricultural services, lack of intelligent decision-support systems, overreliance on intermediaries, and limited accessibility to agricultural expertise. Understanding these weaknesses provides justification for the development of a more integrated and intelligent solution.

Furthermore, the chapter introduces the proposed Green Market system and explains its conceptual design and architectural structure. The proposed platform is designed as a centralized digital ecosystem that integrates an agricultural marketplace, expert consultation services, and AI-powered analytics into a unified web-based application. The conceptual and architectural discussions provide insight into how the system components interact to support secure, scalable, and efficient operations.

The chapter also presents detailed descriptions of the major components within the proposed architecture, explaining the responsibilities and functions of each module, including the marketplace component, authentication system, database management system, AI integration layer, and expert consultation services. Although detailed algorithms and implementation procedures are reserved for Chapter Four, this chapter provides a high-level explanation of how each component operates within the overall system.

Additionally, the proposed software features, development tools, and implementation environment are discussed. Technologies such as Next.js, TypeScript, Tailwind CSS, PostgreSQL, Drizzle ORM, and Supabase are introduced as the core technologies that will support the development of the platform. Finally, the chapter highlights the expected benefits of implementing the Green Market system, particularly in improving market accessibility, enhancing transparency, supporting data-driven agricultural decision-making, and increasing operational efficiency within the agricultural ecosystem.

2.2 Review of Related Works / Review of Similar Systems

Several agricultural technology platforms and digital marketplace systems have been developed to improve agricultural productivity, market accessibility, and farmer support services. Examples of such platforms include Grow For Me, Farmcrowdy, and DeHaat. These systems provide digital solutions that help connect farmers, investors, buyers, and agricultural stakeholders within the agricultural value chain.

Grow For Me mainly focuses on agricultural crowdfunding and investment opportunities, allowing individuals to sponsor farming activities while supporting farmers financially. Farmcrowdy also provides agricultural financing and digital farmer support services through sponsorship and investment models. Similarly, DeHaat offers agricultural services such as market linkage, farm input supply, and advisory support to farmers.

Although these platforms have contributed significantly to agricultural digitalization, many of them focus mainly on specific services such as crowdfunding, financing, or marketplace operations. Most existing systems lack a fully integrated environment that combines marketplace services, expert consultation, and AI-powered analytics within one platform.

In addition, some existing systems provide limited personalization and data-driven recommendations for users. Farmers in rural communities may still experience difficulties accessing verified agricultural experts and reliable market insights. Some platforms also have limited transparency in transactions and communication between stakeholders. These challenges create the need for a more comprehensive and intelligent agricultural platform.

The proposed Green Market system seeks to address these limitations by integrating an agricultural marketplace, expert consultation services, and AI-driven insights into a single web-based platform. This approach is intended to improve efficiency, enhance transparency, support better decision-making, and create a more connected agricultural ecosystem.

2.3 Processes of Existing Systems

Existing agricultural and digital marketplace systems generally support functionalities such as user registration, product listing, online transactions, communication systems, ratings and reviews, and basic farmer support services. Users typically create accounts on the platform and are assigned specific roles such as farmers, buyers, or administrators. After registration, farmers can upload and manage product listings by providing details such as product name, quantity, pricing, and availability.

Buyers interact with the system by browsing available products, searching for specific items, and placing orders through the platform. Most systems also include communication features that allow users to interact with sellers or customer support teams regarding transactions and product inquiries. In addition, rating and review systems are commonly implemented to improve transparency, accountability, and trust between users.

Some agricultural platforms further provide support services such as market price updates, farming tips, financial assistance, and basic advisory services. These services are intended to improve agricultural productivity and market accessibility for farmers. However, many existing

systems operate with limited integration between their various services, resulting in fragmented user experiences and reduced operational efficiency.

Although these systems have improved digital participation in agriculture, several platforms still face challenges related to scalability, intelligent analytics, personalized recommendations, and centralized expert consultation services. These limitations highlight the need for a more integrated and intelligent agricultural platform such as Green Market.

2.3.1 Advantages of Existing Systems

Existing agricultural technology platforms and digital marketplace systems provide several important benefits to stakeholders within the agricultural sector. One major advantage is the improved accessibility to agricultural products and services. Through digital platforms, farmers and buyers can interact regardless of their geographical locations, thereby expanding market reach and reducing barriers to trade.

Another advantage is faster communication between stakeholders such as farmers, buyers, investors, and agricultural service providers. Most platforms include communication and notification systems that enable users to exchange information efficiently and manage transactions more effectively. This improves coordination within the agricultural value chain.

Existing systems also improve market visibility for farmers by allowing them to showcase their products to a wider audience through online marketplaces. This creates more opportunities for sales and increases awareness of available agricultural products. In addition, many systems incorporate ratings and review mechanisms that promote transparency and trust between users by allowing buyers to provide feedback on products and services received.

Furthermore, digital agricultural systems help reduce manual operational processes by automating activities such as product listing, order management, and user registration. This improves efficiency, reduces paperwork, and minimizes delays associated with traditional agricultural trading methods.

2.3.2 Limitations of Existing Systems

Despite the numerous advantages provided by existing agricultural platforms, several limitations still exist. One major limitation is the continued dependence on intermediaries in some systems, which often reduces farmer profitability and creates unfair pricing structures within the agricultural market.

Another significant challenge is the limited access to verified agricultural experts. Many existing systems do not provide structured or centralized consultation services, making it difficult for farmers, especially those in rural areas, to obtain timely professional guidance and technical support.

In addition, most existing systems lack AI-powered insights and predictive analytics that can support intelligent decision-making. Features such as demand forecasting, personalized recommendations, pricing analysis, and market trend predictions are either limited or completely absent in many agricultural platforms.

Many agricultural services are also fragmented across multiple independent systems. Users often need to rely on different platforms for marketplace operations, expert consultation, communication, and analytics, resulting in poor user experience and reduced operational efficiency. Furthermore, there is often poor integration between marketplace systems and consultation services, limiting effective collaboration and information sharing among stakeholders within the agricultural ecosystem.

2.4 The Proposed System

The proposed system, Green Market, is a web-based integrated agricultural platform designed to improve efficiency, accessibility, and transparency within the agricultural sector. The system combines three major services – a digital agricultural marketplace, agricultural expert consultation services, and AI-powered analytics into a single unified ecosystem. The platform is intended to address the major challenges faced by farmers, buyers, and other agricultural stakeholders by leveraging modern web technologies and intelligent digital solutions.

The marketplace component of the system enables farmers and sellers to list agricultural products with detailed descriptions, pricing, and availability information. Buyers can browse products, search for specific items, compare prices, place orders, and interact directly with farmers without excessive dependence on intermediaries. This direct interaction is expected to improve farmer profitability while ensuring fair pricing and better product accessibility for buyers.

The expert consultation module provides users with access to verified agricultural professionals such as agronomists, veterinarians, and crop specialists. Through the platform, users can book consultations, seek professional advice, and receive guidance on farming practices, pest control, crop management, and other agricultural challenges. This feature improves access to agricultural expertise, particularly for farmers in rural and underserved communities.

In addition, the proposed system incorporates AI-powered analytics and intelligent recommendation systems to support data-driven decision-making. The AI module provides features such as market trend analysis, demand prediction, pricing insights, sales analytics, and personalized recommendations for users. The platform also includes an AI chatbot that assists users by providing quick responses to common agricultural questions and guidance.

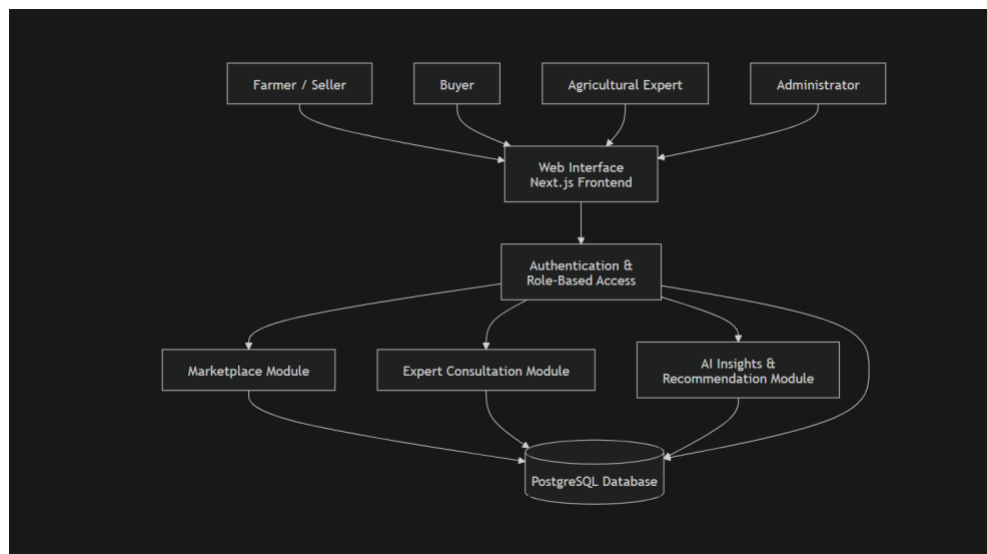
The Green Market platform is designed with a role-based access structure that supports different categories of users including farmers, buyers, agricultural experts, and administrators. Each user category is provided with functionalities and permissions that align with their responsibilities within the system. Furthermore, the system is designed to be responsive, scalable, secure, and user-friendly, ensuring efficient performance across different devices and user environments.

Overall, the proposed system seeks to create a centralized digital agricultural ecosystem that enhances market accessibility, improves communication between stakeholders, promotes intelligent agricultural practices, and contributes to sustainable agricultural development.

2.5 Conceptual Design

The conceptual design of Green Market illustrates how farmers, buyers, agricultural experts, and administrators interact with the platform through a web interface. Farmers can list products, manage orders, and access expert advice, while buyers can browse products, place orders, and receive recommendations. Agricultural experts provide consultation services, and administrators manage the overall system activities. The web interface communicates with backend services responsible for authentication, marketplace operations, consultation management, AI analytics, and database processing. The system database stores user information, products, orders, consultations, and analytics data, while the AI module provides intelligent insights such as recommendations, demand prediction, and chatbot assistance. Overall, the conceptual design presents Green Market as a centralized digital agricultural ecosystem that integrates marketplace services, expert consultation, and AI-powered decision support into one unified platform.

Figure 2.1: Conceptual Design of Green Market

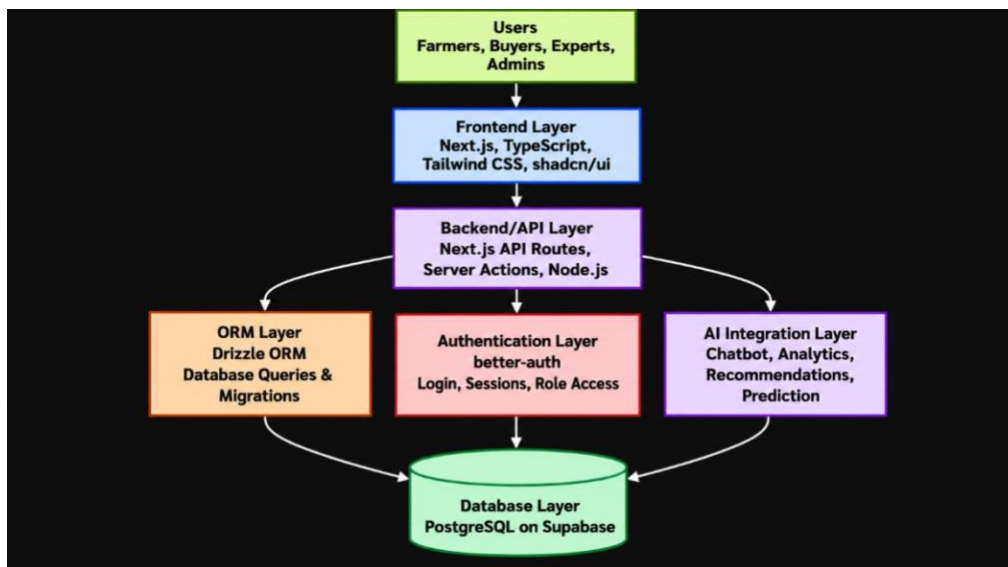


This conceptual design demonstrates that Green Market operates as a centralized digital agricultural ecosystem. All users interact with the system through the web interface, while the backend components manage authentication, marketplace operations, expert consultation services, AI-powered recommendations, and database management. The integration of these components ensures efficient communication, secure data processing, and smooth agricultural operations within the platform.

2.6 Architecture of the Proposed System

The Green Market platform follows a layered full-stack architecture consisting of a frontend layer, backend/API layer, authentication layer, ORM layer, database layer, and AI integration layer. The architecture is designed to ensure scalability, security, maintainability, and good system performance. Users interact with the system through the frontend interface built with Next.js, TypeScript, Tailwind CSS, and shadcn/ui. The frontend sends user requests to the backend/API layer, where business logic such as product management, order processing, expert booking, and AI-related requests are handled. The authentication layer manages login, registration, user sessions, and role-based access control for farmers, buyers, experts, and administrators. The ORM layer, implemented using Drizzle ORM, connects the backend to the PostgreSQL database hosted on Supabase. The AI integration layer supports intelligent features such as chatbot assistance, demand prediction, pricing insights, and personalized recommendations.

Figure 2.2: Architecture of the Proposed System



This architecture demonstrates how the major system layers of Green Market interact to support efficient and secure agricultural operations. The frontend layer allows users to interact with the platform, while the backend/API layer processes requests and manages business logic. The authentication layer ensures secure user access and role-based authorization, whereas the ORM and database layers handle data storage, retrieval, and management. In addition, the AI integration layer provides intelligent features such as chatbot assistance, recommendations, analytics, and demand prediction. Together, these layers create a scalable, secure, and reliable agricultural platform that supports smooth marketplace activities, expert consultation services, and data-driven decision-making.

2.7 Components Designs and Components Descriptions

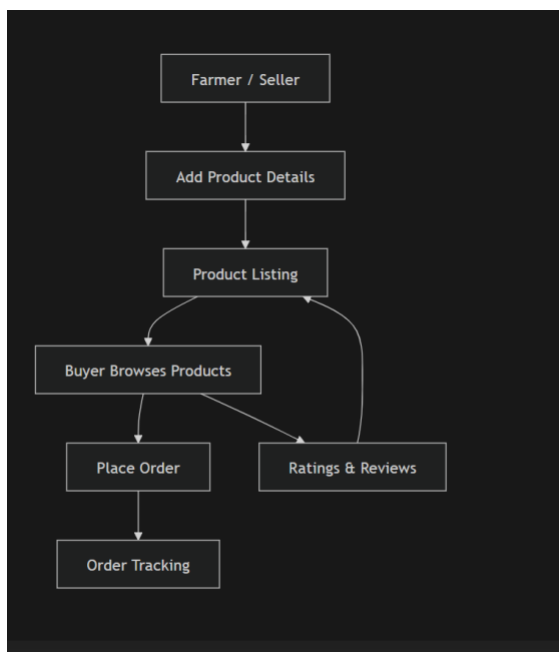
The Green Market system consists of several major components that work together to ensure efficient and reliable platform operations. These components include the marketplace module, expert consultation module, authentication and authorization system, AI recommendation engine, database management system, and API communication layer. Each component performs specific responsibilities within the system architecture and contributes to the overall functionality of the platform.

2.7.1 Marketplace Module

The marketplace module serves as the core trading component of the system. It allows farmers and sellers to upload agricultural products with details such as product name, pricing, quantity, category, description, and availability. Buyers can browse available products, search and filter items based on their preferences, compare prices, and place orders through the platform.

The module also manages important marketplace operations such as order processing, product updates, ratings, reviews, and transaction tracking. Farmers are able to manage inventory and monitor product performance, while buyers can track purchases and provide feedback on products received. The inclusion of ratings and reviews improves trust, transparency, and accountability between users within the platform. In addition, the marketplace module interacts with the authentication system to verify users and ensure that only authorized sellers can upload or manage products. The component also communicates with the database system to store product information, order records, and transaction details securely. Overall, this module improves accessibility, reduces dependence on intermediaries, and creates a more efficient digital agricultural marketplace.

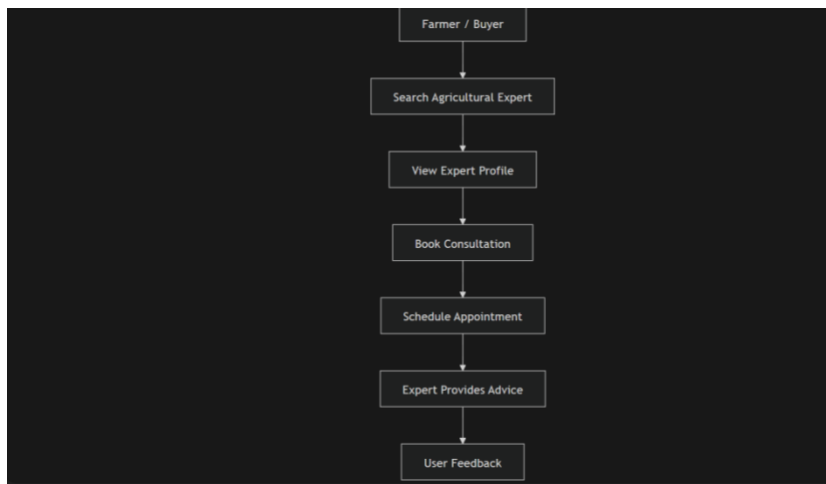
Figure 2.3: Marketplace Module Design



2.7.2 Expert Consultation Module

The expert consultation module connects users with verified agricultural experts such as agronomists and veterinarians. Users can book consultations, communicate with experts, and receive professional agricultural guidance. The module manages consultation scheduling, expert profiles, bookings, and communication processes. This improves access to agricultural expertise, especially for rural farmers who may have limited access to professional support.

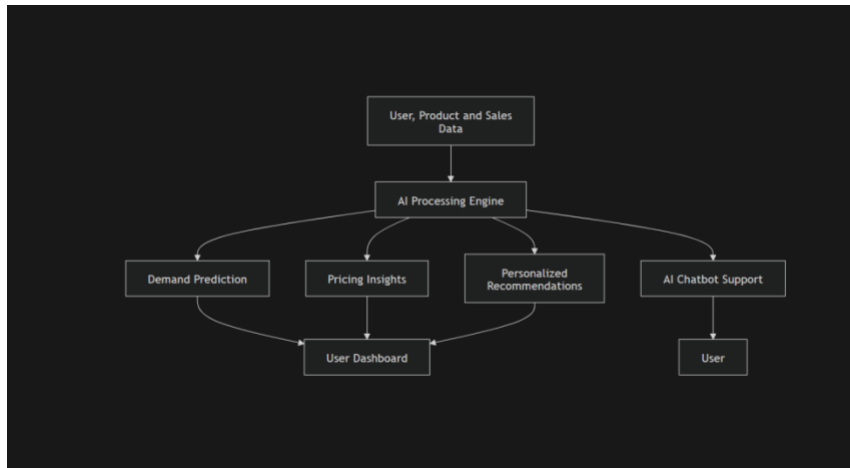
Figure 2.4: Expert Consultation Module Design



2.7.4 AI Recommendation Engine

The AI recommendation engine provides intelligent insights and decision-support services within the platform. The component processes user activities, sales records, and market data to generate personalized recommendations, pricing insights, and demand predictions. The AI module also powers the chatbot system, which provides users with basic agricultural assistance and quick responses to common questions.

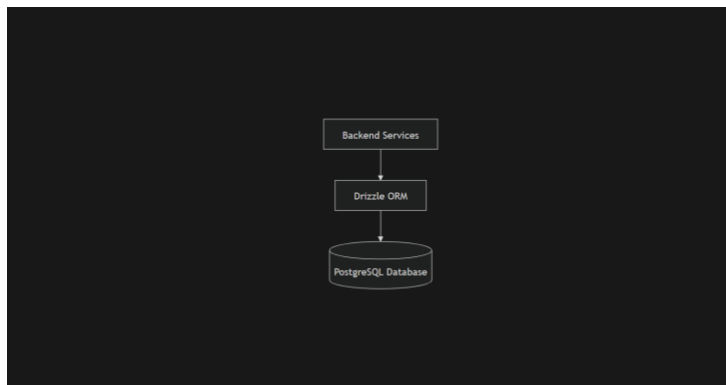
Figure 2.6: AI Recommendation Engine Design



2.7.5 Database Management System

The database management component stores and manages all system data including user accounts, products, orders, consultations, reviews, and analytics records. The system uses PostgreSQL hosted on Supabase for reliable and scalable data storage. Drizzle ORM is used to manage database queries, relationships, and migrations efficiently. This component ensures data consistency, security, and efficient retrieval of information.

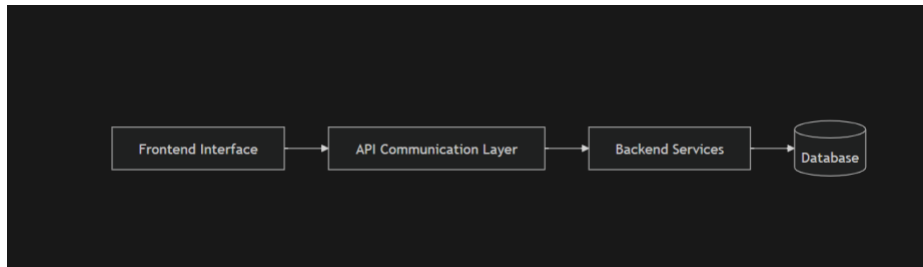
Figure 2.7: Database Management System Design



2.7.6 API Communication Layer

The API communication layer enables interaction between the frontend interface and backend services. This component processes requests from users, communicates with the database and AI modules, and returns responses to the frontend. The API layer handles operations such as product management, authentication, consultation bookings, analytics processing, and order management. It plays an important role in ensuring smooth communication across all system layers.

Figure 2.8: API Communication Layer Design



Together, these components form the core structure of the Green Market platform and ensure smooth, secure, and efficient agricultural operations within the system.

2.8 Proposed System/Software Features

The Green Market system will include several features designed to improve agricultural trade, expert support, and intelligent decision-making. These features are developed to support the different roles within the platform, including farmers, buyers, agricultural experts, and administrators. The system will provide user registration and authentication to allow users to securely create accounts and access the platform. It will also include role-based access control to ensure that each user can only access features relevant to their role. For example, farmers can manage products, buyers can place orders, experts can manage consultations, and administrators can monitor the system. The platform will also include product listing and management features that allow farmers to upload agricultural products with details such as product name, price, quantity, description, and availability. Buyers will be able to browse, search, and filter products before placing orders. In addition, the system will support expert consultation booking, allowing users to schedule sessions with verified agricultural professionals. Another important feature is the AI chatbot, which will provide basic agricultural guidance and quick responses to common user questions. The system will also include a sales analytics dashboard, demand prediction system, pricing insights, and personalized recommendations to help users make better decisions. Ratings and reviews will be included to promote trust and transparency between users. Finally, the system will be designed with responsive interfaces to ensure usability across desktops, tablets, and mobile devices. Secure database management will also be implemented to protect user data, product records, orders, consultations, and analytics information. Overall, these features will make Green Market a reliable, intelligent, and user-friendly agricultural platform.

2.9 Development Tools and Environment

The development of the Green Market platform will utilize modern web development tools and technologies to ensure scalability, security, maintainability, and efficient system performance. The system will be developed using Next.js with TypeScript for both frontend and backend development. Next.js provides a full-stack development environment that supports server-side rendering, API routes, routing, and efficient web application performance. For the user interface design, Tailwind CSS and shadcn/ui will be used to create responsive, modern, and user-friendly interfaces. These tools will help improve the visual appearance, responsiveness, and usability of the platform across different devices including desktops, tablets, and mobile phones. The database system for the project will be PostgreSQL hosted on Supabase. PostgreSQL will

provide reliable and scalable relational database management for storing user records, product information, consultations, orders, reviews, and analytics data. Supabase will serve as the cloud database hosting platform and backend service provider. Database interactions and schema management will be handled using Drizzle ORM together with drizzle-kit for migrations and database version control. This will allow efficient query management, type-safe database operations, and easier maintenance of database structures throughout development.

Authentication and authorization functionalities will be implemented using better-auth to support secure login, user registration, session management, and role-based access control for farmers, buyers, experts, and administrators. In addition, GitHub will be used for version control and collaboration, while Visual Studio Code will serve as the primary development environment for coding, debugging, and project management.

Overall, these tools and technologies provide a modern and efficient development environment capable of supporting the successful implementation of the Green Market platform.

2.10 Benefits of Implementation of the Proposed System

The implementation of the Green Market platform is expected to provide several important benefits to stakeholders within the agricultural sector. One major benefit is the improvement of farmer profitability through direct interaction between farmers and buyers. By reducing excessive dependence on intermediaries, farmers can sell products at fair prices and gain better access to larger markets. The system will also enhance market accessibility by providing a centralized digital marketplace where buyers can easily search, browse, and purchase agricultural products online. This improves product visibility and creates more business opportunities for farmers and agricultural businesses. Another important benefit is easier access to agricultural expertise. Through the expert consultation module, farmers and buyers can connect with verified agricultural professionals for advice on farming practices, pest control, crop management, and other agricultural challenges. This can help improve productivity and reduce farming risks. In addition, the integration of AI-powered analytics and recommendation systems will support data-driven decision-making. Features such as demand prediction, pricing insights, personalized recommendations, and sales analytics will help users make informed agricultural and business decisions. The proposed system will also improve transparency and accountability through ratings, reviews, and transaction tracking features. Automated digital processes such as product management, order processing, and consultation booking will reduce operational inefficiencies and minimize manual work. Furthermore, the platform will promote better communication and collaboration among farmers, buyers, experts, and administrators within the agricultural ecosystem. Overall, the implementation of Green Market is expected to contribute to sustainable agricultural growth, improved efficiency, enhanced productivity, and increased digital transformation within the agricultural sector.

CHAPTER THREE

METHODOLOGY

3.1 Chapter Overview

This chapter presents the methodology used in the design and development of the Green Market platform. It discusses the system requirements, stakeholders involved, requirement gathering process, functional and non-functional requirements, UML diagrams, security concepts, software process models, project design considerations, database design, user interface design, and development tools used for the implementation of the proposed system. The chapter also explains the selected software development methodology and provides justification for its adoption in the development of the Green Market platform. In addition, various system models and diagrams are presented to illustrate the structure, behavior, and interactions within the system.

3.2 Requirement Specification

Requirement specification refers to the process of identifying, analyzing, and documenting the functionalities and constraints of the proposed system. The requirement specification for Green Market was developed to ensure that the system satisfies the needs of farmers, buyers, agricultural experts, and administrators. The requirements were grouped into functional requirements and non-functional requirements to clearly define the operations, performance expectations, security needs, and usability standards of the platform.

3.3 Stakeholders of the System

The Green Market platform consists of several stakeholders who interact with the system directly or indirectly to ensure efficient agricultural operations and smooth platform activities. These stakeholders include farmers/sellers, buyers, agricultural experts, and administrators, each performing specific responsibilities within the system.

3.3.1 Farmers/Sellers

Farmers and sellers are primary users of the platform who use the system to upload and manage agricultural products, monitor inventory, track orders, manage sales activities, and interact directly with buyers. They can also access AI-powered insights, market recommendations, and consultation services from agricultural experts to improve productivity and decision-making. The platform helps farmers increase market reach and reduce dependence on intermediaries.

3.3.2 Buyers

Buyers interact with the platform to browse available agricultural products, search and filter items, place orders, track purchases, and provide ratings and reviews. Buyers can also receive personalized recommendations and market insights generated by the AI recommendation system. The platform improves accessibility to agricultural products and promotes transparent transactions between buyers and sellers.

3.3.3 Agricultural Experts

Agricultural experts provide professional consultation services and agricultural guidance to users of the platform. These experts may include agronomists, veterinarians, crop specialists, and other agricultural professionals. They manage consultation appointments, respond to inquiries, and provide recommendations on farming practices, pest control, crop management, and other agricultural challenges.

3.3.4 Administrators

Administrators are responsible for managing and monitoring the overall operations of the Green Market platform. Their duties include managing users, verifying experts, monitoring transactions, generating reports, maintaining system security, and ensuring smooth platform performance. Administrators help enforce system policies and ensure that the platform operates efficiently and reliably.

3.4 Requirement Gathering Process

The requirement gathering process involved collecting information from potential users and analyzing existing agricultural systems to identify the major challenges within the agricultural ecosystem.

The following techniques were used:

- Observation of existing agricultural marketplace systems.
- Review of related agricultural platforms such as Grow For Me, Farmcrowdy, and DeHaat.
- Informal discussions with students, farmers, and online agricultural communities.
- Analysis of agricultural business processes and digital marketplace operations.

The information gathered helped identify the major system requirements and functionalities needed for the development of Green Market.

3.5 Functional Requirements

The functional requirements describe the major operations and services the system must provide.

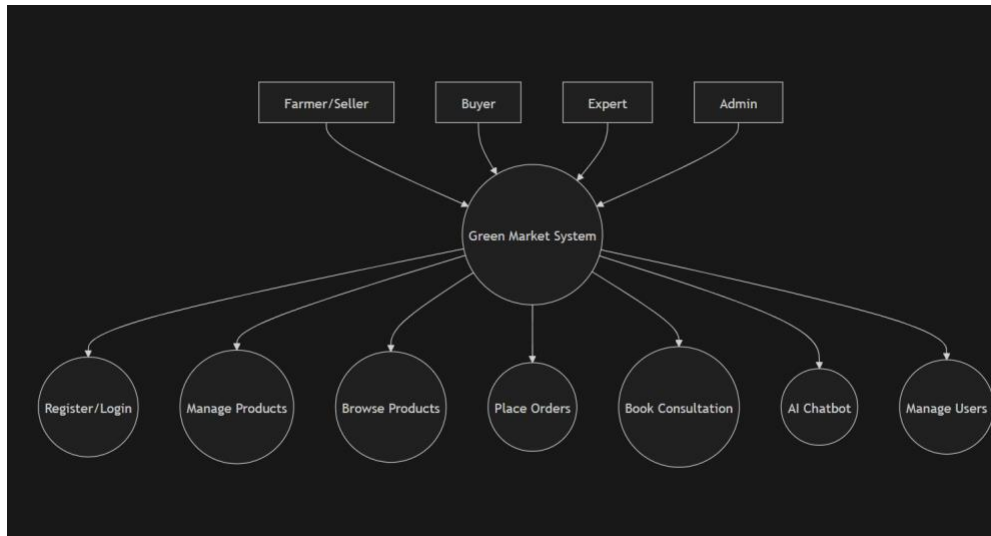
The Green Market platform should be able to:

- Allow users to register and log into the system.
- Support role-based access control.
- Allow farmers to upload and manage products.
- Allow buyers to browse and purchase products.
- Allow users to book consultations with experts.
- Provide AI chatbot assistance.
- Generate analytics and recommendations.
- Allow users to rate and review products.
- Manage orders and transactions.
- Store and retrieve system data securely.

3.6 UML Diagrams

UML diagrams are used to visually represent the structure and behavior of the Green Market system.

3.6.1 Use Case Diagram (Frontend Models)



3.6.2 Use Case Description

Farmer

The farmer uploads products, manages orders, books consultations, and receives AI recommendations.

Buyer

The buyer browses products, places orders, tracks purchases, and interacts with the AI chatbot.

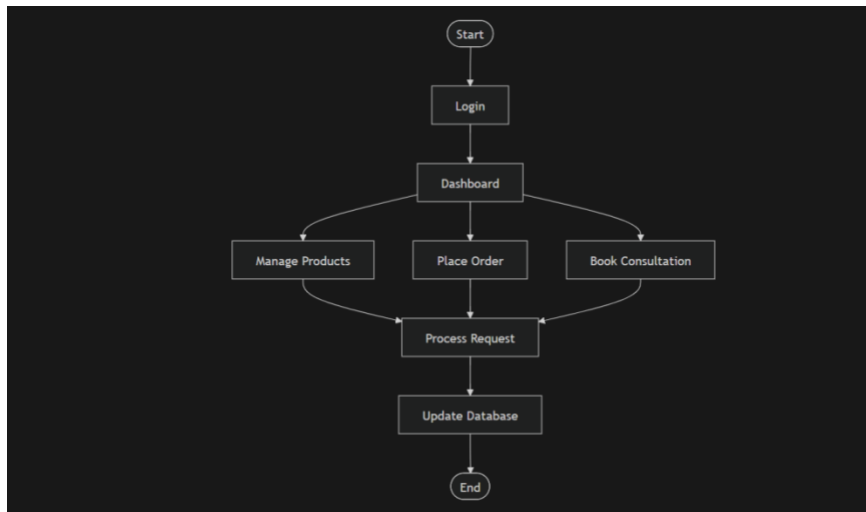
Expert

The agricultural expert manages consultation appointments and provides agricultural advice.

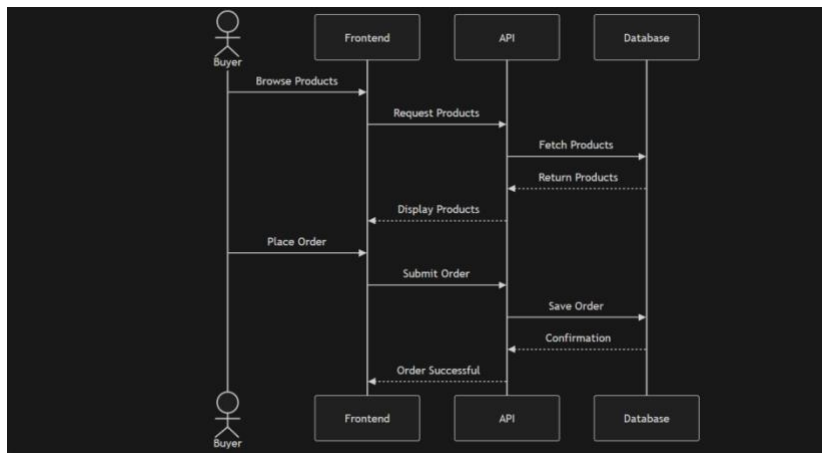
Administrator

The administrator manages users, reports, and system monitoring activities.

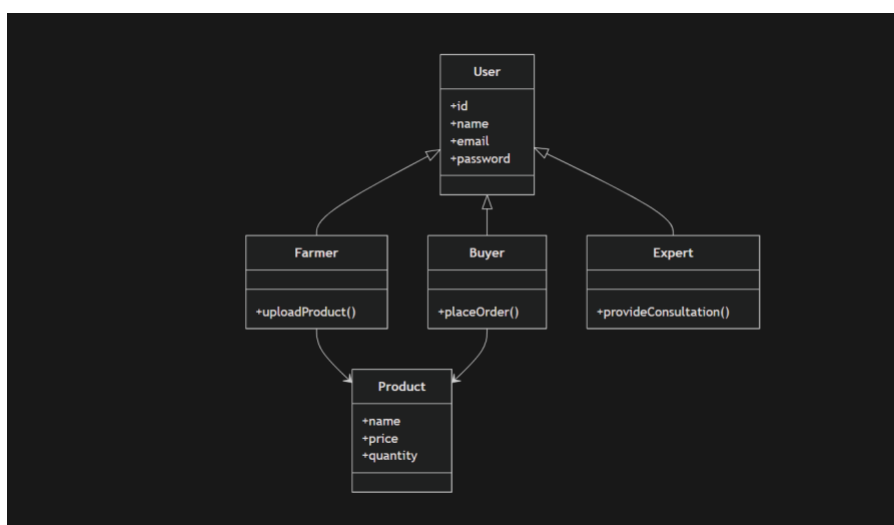
3.6.3 Activity Diagram



3.6.4 Sequence Diagram



3.6.5 Class Diagram



3.7 Non-Functional Requirements

Non-functional requirements describe the quality standards, performance expectations, and operational characteristics of the Green Market platform. These requirements define how the system should perform rather than the specific functionalities it provides. The non-functional requirements are important in ensuring that the platform remains secure, reliable, scalable, efficient, and user-friendly for all stakeholders.

3.7.1 Performance

The Green Market platform should provide fast and efficient processing of user requests and system operations. The system must be capable of handling multiple users simultaneously without significant delays or reduction in performance. Activities such as user authentication, product browsing, order placement, consultation booking, and AI recommendation generation should execute within reasonable response times to ensure a smooth user experience.

The use of modern technologies such as Next.js, PostgreSQL, and Drizzle ORM is expected to improve overall application performance and optimize data processing operations. Efficient database queries and API communication will also help minimize latency and improve system responsiveness.

3.7.2 Security

Security is a critical requirement of the Green Market platform because the system manages sensitive user information, transactions, consultation records, and marketplace activities. The system should provide secure authentication, encrypted sessions, protected database access, and secure communication between all system components.

The platform will implement role-based access control to ensure that users can only access functionalities permitted for their specific roles. Authentication and session management will be handled securely using better-auth. Passwords and sensitive credentials will be encrypted to reduce the risk of unauthorized access and data breaches.

Furthermore, secure APIs, input validation, and controlled database interactions will help protect the platform against malicious attacks such as SQL injection, unauthorized access, and data manipulation.

3.7.3 Scalability

The Green Market system should be scalable enough to support future expansion and increasing numbers of users, products, consultations, and transactions. The platform architecture is designed to accommodate growth without requiring major structural changes to the system.

The layered full-stack architecture, cloud-based database hosting on Supabase, and modular system design allow additional features and services to be integrated easily in the future. This

scalability ensures that the platform can continue to perform effectively as user demand increases over time.

3.7.4 Usability

The platform should provide a responsive, user-friendly, and intuitive interface that is easy to understand and operate. Users should be able to navigate the system efficiently regardless of their technical background or level of digital literacy.

The interface will be designed using Tailwind CSS and shaden/ui to provide a clean, modern, and responsive user experience across desktops, tablets, and mobile devices. Proper layout organization, clear navigation menus, and accessible interface elements will improve overall usability and user satisfaction.

3.7.5 Reliability

The Green Market platform should provide reliable and stable operations with minimal downtime or system failures. The system should consistently perform expected operations such as authentication, product management, order processing, consultation booking, and analytics generation without interruption.

Reliable database management, secure cloud hosting, and proper error handling mechanisms will help maintain system stability and data integrity. Backup and recovery mechanisms will also help protect important system data against accidental loss or corruption.

3.8 Security Concepts

The Green Market platform implements several security mechanisms to protect user data, transactions, and overall system operations. Since the platform handles sensitive information such as user accounts, product transactions, consultations, and AI-generated insights, proper security measures are necessary to maintain confidentiality, integrity, and system reliability.

One of the primary security mechanisms implemented in the system is user authentication and authorization using better-auth. This ensures that only verified users can access the platform and perform authorized activities based on their assigned roles. Farmers, buyers, agricultural experts, and administrators are each provided with specific permissions and access rights within the system.

The system also implements password encryption and secure session management to protect user credentials and maintain secure login sessions. Sensitive user information is encrypted before storage to reduce the risk of credential theft and unauthorized access.

In addition, role-based access control is used to restrict users from accessing functionalities outside their assigned responsibilities. For example, only administrators can manage users and system reports, while only experts can manage consultation services.

Protected database access is implemented using secure APIs and controlled backend communication to prevent unauthorized interaction with the PostgreSQL database. The API layer validates requests before processing them to ensure data security and consistency.

The platform further implements input validation and sanitization techniques to reduce malicious attacks such as SQL injection, cross-site scripting, and invalid data submission. All user inputs are checked and validated before being processed by the backend system.

Secure communication between the frontend and backend services is also maintained through protected API requests and secure session handling. These security mechanisms collectively help protect the system against unauthorized access, data breaches, cyber threats, and malicious activities while ensuring safe and reliable platform operations.

3.9 Project Methods

Software development methodologies provide structured approaches for designing and developing software systems. The major software process models include the Waterfall Model, Spiral Model, Prototype Model, Agile Model, and Incremental Model. Each model provides different approaches for system planning, development, testing, and maintenance.

3.10 Chosen Model and Justification

The Agile software development model was selected for the development of Green Market. Agile supports iterative development, continuous testing, flexibility, and regular feedback throughout the project lifecycle.

The model was chosen because the Green Market platform contains multiple integrated modules such as the marketplace system, expert consultation services, AI features, and analytics dashboard. Agile makes it easier to develop and improve these modules gradually while identifying and fixing issues during development.

3.11 Project Design Considerations (Logical Designs)

Project design considerations describe the logical structure and visual organization of the Green Market platform.

3.11.1 UI Design

The user interface design focuses on simplicity, responsiveness, accessibility, and good user experience across different devices.

The main UI wireframes include:

- Login and Registration Pages
- Farmer Dashboard
- Buyer Dashboard

- Product Listing Pages
- Consultation Booking Interface
- Analytics Dashboard
- Admin Management Pages

3.11.2 DB Design

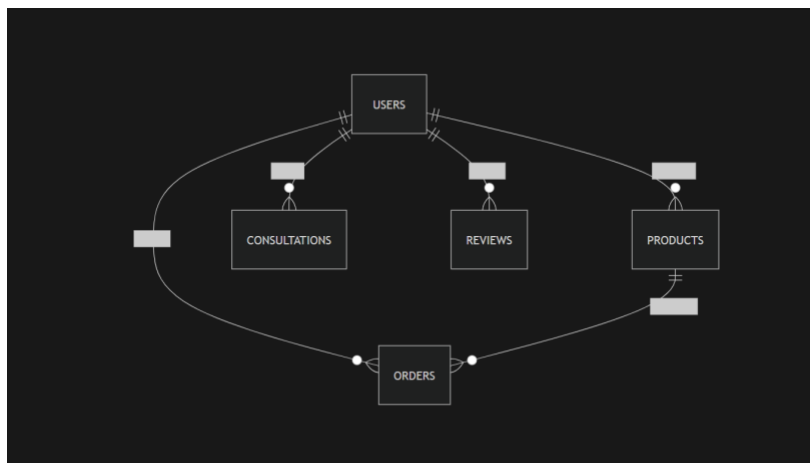
The Green Market database design consists of several related entities used to manage system data efficiently.

The major database entities include:

- Users
- Products
- Orders
- Consultations
- Reviews
- Analytics

The database is implemented using PostgreSQL and managed through Drizzle ORM for efficient and secure database operations.

E-R Diagram



3.12 Developmental Tools

- **Next.js** — Used for building both the frontend interface and backend API services.
- **TypeScript** — Used to improve code quality, readability, and maintainability.
- **Tailwind CSS** — Used for creating responsive and modern user interfaces.
- **shadcn/ui** — Used for reusable and consistent UI components.
- **PostgreSQL** — Used for secure and reliable relational database management.
- **Supabase** — Used for cloud database hosting and backend support services.

- **Drizzle ORM** — Used for database queries, schema management, and migrations.
- **better-auth** — Used for secure authentication and role-based access control.
- **GitHub** — Used for version control, collaboration, and project backup.
- **Visual Studio Code** — Used as the primary environment for coding, debugging, and testing.